

Missing Time

Teacher and presenter guide - bounded rock-record investigation - Murray State University

Purpose and learning target

Students compare surviving rock thickness in metres with elapsed model time in Myr since start. They use labelled synthetic chronology anchors, not thickness, to make a local missing interval visible and limit claims to evidence.

Target claim: a local gap means rock from an interval is absent here. It does not by itself identify cause, prove an extinction boundary, or turn a local fossil absence into a global one.

One Myr is one million years in this authored model. Metres record surviving rock height. Sampling reveals specimens only in surviving rock and never alters the record.

Current app flow

- Select Case 1, 2, or 3 and inspect surviving rock; the time mapping and gaps remain hidden at first.
- Sample surviving layers; the sample ledger stays visible.
- Select **Show supplied age evidence**. Only then do supplied base/top ages for every surviving layer, the assumed within-layer linear mapping, and any missing-interval display appear.
- Choose a warranted conclusion, then reveal the authored history.

Controls: ? reopens how-to. **Settings** contains presentation mode, openable presenter notes, and reset. This app does not calculate time from thickness.

Three authored investigations

Case	Evidence task	Warranted conclusion
1. Section A	Sample middle and top rock, then show supplied ages for A's surviving layers.	No missing interval is represented under the supplied chronology and within-layer continuity assumption; a local last observation does not establish global extinction.
2. Section B	Show supplied ages and sample Red siltstone adjacent below the gap plus Cross-bedded sand above it.	A visible local gap does not identify its cause or establish an extinction boundary.
3. Sections C and D	Show supplied chronology at both C and D; sample below C's gap and D's Green mudstone.	D preserves material from the interval missing at C. A blank is not a global fossil absence.

Chronology evidence and units

Supplied synthetic base and top ages for every surviving layer establish the displayed model-time placement, under an explicit constant-rate linear mapping within each surviving layer. Fossil times are mapped from that supplied chronology; they are not independently measured dates. The left view is surviving height above base in metres; the time view is model Myr since start. Equal thickness does not mean equal elapsed time, and an unsampled layer is not an observed fossil absence.

Scope of the mapping: dating uncertainty and unresolved internal hiatuses are omitted. Case 3 uses Section C ages bracketing C's pause and Section D ages at the base and top of D's Green mudstone. C and D carry the comparison; do not infer the interval from column height.

Authored model and reveal boundaries

The model deposits layers at stated constant rates, can pause deposition, erodes top-down, then resumes deposition. Specimens stay at authored positions only while their rock survives. A removed specimen is not moved to a hiatus boundary.

Case	Authored reveal	Teaching point
1	Asteria's synthetic lifetime ends at 3.40 Myr, inside Green mudstone.	Hidden truth does not make a local sample proof of global extinction.
2	Erosion removes 15.00 m of Pale limestone and 10.00 m from Red siltstone.	Removed fossils are absent from surviving rock.
3	C has a 2.00 Myr pause while D accumulates Green mudstone.	A specimen at D was never moved into C's hiatus.

This is a small authored preservation model, not a basin simulator. It omits reworking, transport, dating uncertainty, sampling statistics, facies correlation, and global distribution.

Suggested facilitation

- Ask what observation could distinguish a local disappearance from extinction.
- Have students name chronology anchors before interpreting a gap.
- For Case 2, point out that erosion removes top-down, including specimens in removed deposits.
- For Case 3, ask what the second local section changes and what it cannot prove.

Conclusion feedback is not a mastery score. It only reports whether the selected statement follows from visible authored evidence.

Source context and attribution

StratPal, Event data, for occurrence-event and preservation context. This activity retains authored synthetic specimens in surviving rock; it does not treat event-data transformations as a specimen-preservation algorithm.

Created by Bryant Harrison - Murray State University. The source is teaching context, not a current assignment or endorsement.